The axiom of choice implies the law of excluded middle

The Univalent Foundations Program

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Editorial note. The counted statement is the author thm environment at source lines 614-619 and its proof environment opens at line 621 after a blank line; the pairing is by adjacency and the retained proof range 621-646 is the whole environment.

Editorial note. The retained counter values (chapter 10, section 1, theorem offset 13) make the counted theorem print as Theorem 10.1.14, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel thm:1surj_to_surj_to_pem = 10.1.14).

Editorial note. Two labels the proof cites lie outside the card and are declared in external_references as non-load-bearing navigation pointers: thm:ac-epis-split (book Lemma 3.8.2, defined at line 742 of the logic chapter pinned file) and prop:trunc_of_prop_is_set (book Lemma 10.1.13, defined at line 537 of this chapter pinned file).

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the axiom of choice implies the law of excluded middle (author label thm:1surj_to_surj_to_pem, book Theorem 10.1.14).

Allowed standard prerequisites (editorial).

- The axiom of choice used here. The form of choice in force says that every surjection between sets has a section.. It is the hypothesis, and the proof recalls its equivalent form.
- Suspension of a mere proposition. The suspension of a mere proposition is a set, and its two poles are equal exactly when the proposition holds.. It is the type on which the argument is run.
- Decidable equality of the booleans. Equality of the two booleans is decidable, so any proposition obtained by transporting it is decidable too.. It produces the disjunction at the end.

Source reference. Original lemma 3.8.2: sources/hott-logic/pinned-original-source.tex, source lines 742–742. Book Lemma 3.8.2, the equivalent form of the axiom of choice used in the retained proof, cited at source line 622. Navigation pointer only: the form in force is restated in the retained text.

Source reference. Original lemma 10.1.13: sources/hott-setmath/pinned-original-source.tex, source lines 537–537. Book Lemma 10.1.13, that the suspension of a mere proposition is a set and that its poles are equal exactly when the proposition holds, cited twice in the retained proof. Navigation pointer only.

Theorem 10.1.14 (Diaconescu)**.** *The axiom of choice implies the law of excluded middle.*

Proof. We use the equivalent form of choice given in original lemma 3.8.2. Consider a mere proposition A. The function $f : \mathbf{2} \to \Sigma(A)$ defined by $f(0_{\mathbf{2}}) :\equiv \mathsf{N}$ and $f(1_{\mathbf{2}}) :\equiv \mathsf{S}$ is surjective. Indeed, we have $(0_{\mathbf{2}}, \mathsf{refl}_{\mathsf{N}}) : \mathsf{fib}_f(\mathsf{N})$ and $(1_{\mathbf{2}}, \mathsf{refl}_{\mathsf{S}}) : \mathsf{fib}_f(\mathsf{S})$. Since $\left\|\mathsf{fib}_f(x)\right\|$ is a mere proposition, by induction the claimed surjectivity follows.

By original lemma 10.1.13 the suspension $\Sigma(A)$ is a set, so by the axiom of choice there merely exists a section $g : \Sigma(A) \to \mathbf{2}$ of f. As equality on $\mathbf{2}$ is decidable we get

$$(g(f(0_{\mathbf{2}})) = g(f(1_{\mathbf{2}}))) + \neg(g(f(0_{\mathbf{2}})) = g(f(1_{\mathbf{2}}))),$$

and, since g is a section of f, hence injective,

$$(f(0_{\mathbf{2}}) = f(1_{\mathbf{2}})) + \neg(f(0_{\mathbf{2}}) = f(1_{\mathbf{2}})).$$

Finally, since $(f(0_{\mathbf{2}}) = f(1_{\mathbf{2}})) = (\mathsf{N} = \mathsf{S}) = A$ by original lemma 10.1.13, we have $A + \neg A$. □